LUKE R. HORTON

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EDUCATION

Aerospace Engineering, B.S. | San Diego State University **Aug. 2022 - May 2026**

- **Coursework:** Spacecraft Attitude Dynamics & Control, Feedback Flight Control, Astrodynamics, Rocket & Space Propulsion, Aircraft Stability & Control, Programming & Numerical Methods

EXPERIENCE & PROJECTS

True Anomaly **Denver, CO**

Program Manager Intern *June 2026 - Present*

- Leading my intern project across the company's active defense-space programs: a comprehensive PMO and Engineering template library driving program lifecycle execution and design reviews company-wide.
- Built templates for major lifecycle milestones and review gates (SRD, ICD, PDR/CDR/TRR, IBR, WBS), defining entry/exit criteria and required artifacts for each gate, benchmarked against DoD 5000, NASA SE, and INCOSE standards.
- Designed and shipped an enterprise-shareable AI tool that auto-populates milestone review decks from a single program input, replacing manual slide-by-slide edits and cutting processing cost ~94%.
- Provide support across the company's spacecraft programs wherever needed, engaging cross-functionally with engineering, production/test, and software teams to understand program execution end to end.

Daedalus – Custom Flight Computer & Test Infrastructure | Personal Project *June 2026 – Present*

- Built a custom high-power rocketry flight computer from scratch: C++17 firmware (PlatformIO) on a Teensy 4.1 with ICM-20948 IMU, BMP390 barometer, and uBlox NEO-M9N GPS, a complementary-filter state machine, MOSFET pyro outputs, and 100 Hz SD logging.
- Engineered a full test pipeline: a Software-in-the-Loop (SITL) framework with a 50+ scenario Python pytest suite in GitHub Actions CI, plus a Hardware-in-the-Loop (HITL) bench with automated execution and fault injection (sensor dropout, low battery, pyro continuity loss).
- Earned Level 1 high-power rocketry certification, flying the board against a commercial reference altimeter and validating apogee and flight data in post-flight analysis. Public GitHub repo and writeup.

Ground Effect Vehicle (GEV) **San Diego State University, Senior Design**

Project Manager (Avionics & Systems Lead) *Spring 2026*

- Led a 4-person design-build-fly team delivering an RC-scale GEV in one semester: a 3D-printed airframe on a high-lift airfoil with distributed 4-motor propulsion to capture in-ground-effect performance (speed, ride height, power). Flew successfully in ground effect; showcased at SDSU Senior Design Day.
- As Project Manager, owned the schedule and scope to hit a single-semester design-build-fly-test (DBFT) timeline; led all avionics and flight software, troubleshooting control-surface mixing and full-system integration and tuning fly-by-wire (FBW) autopilot flight modes through iterative flight testing.
- Owned avionics integration: configured a Pixhawk 6C running ArduPlane, an ExpressLRS 2.4GHz CRSF link (RadioMaster Boxer TX, RP1 receiver), mapped 7 control surfaces to 6 FMU PWM channels via Y-splitter distribution, and set up a PM07 module for 6S battery monitoring with voltage and current failsafes.

Next-Generation Carrier-Based Strike Fighter **San Diego State University, Senior Capstone**

Performance Lead & Systems Engineer *Fall 2025 - Spring 2026*

- Led performance and test-planning analysis for a conceptual Navy carrier-based strike fighter (AIAA RFP): translated mission and carrier requirements into verification metrics, owned weight estimation, wing sizing, and takeoff/landing analyses, built physics-based MATLAB sizing models (drag polar, T/W–W/S trades, constraint sets), and delivered technical memos at PDR and CDR.
- Ran structured parametric sweeps and sensitivity checks to quantify carrier-suitability margins, then refined and locked the configuration for CDR based on integrated subsystem feedback.

LEADERSHIP

President, *Kappa Alpha Order* *2024*

- Led a 170+ member chapter and $200K+ budget, directing an executive board and 20+ chairs; earned IFC Most Improved Chapter (2024) for measurable gains in membership and engagement.

SKILLS

Avionics & Embedded: Pixhawk 6C, ArduPilot, Mission Planner, MAVLink, Teensy 4.1, C++17/PlatformIO, IMU/barometer/GPS integration, MOSFET pyro outputs, ELRS/SiK telemetry

Test, Simulation & Programming: SITL/HITL, fault injection, pytest, GitHub Actions (CI), Python, C/C++, Bash/CLI, Git, Linux/SSH, MATLAB, Simulink, ANSYS

PMO & PLM Tools: Jira, Confluence, Duro, Teamcenter, PowerBI, Microsoft Office, Google Suite

CAD & Fabrication: SolidWorks, GD&T, Design for Manufacturability, 3D printing, MIG/TIG welding, machining, soldering & electronics assembly